Shashank Madipelly

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PROFESSIONAL SUMMARY

GenAI Engineer specializing in production LLM systems — RAG pipelines, multi-agent workflows, and FastAPI microservices built on LangChain, LangGraph, and OpenAI/Anthropic APIs. Strength in evaluation and observability: RAGAS, TruLens, and LangSmith frameworks that catch hallucinations before production and track cost, latency, and accuracy together. Partners with product and engineering stakeholders to turn evaluation results into shipped reliable systems.

SKILLS

GenAI & LLMs: Prompt Engineering, Fine-Tuning, GPT-4o, Claude, Gemini, Llama
Frameworks: LangChain, LangGraph, CrewAI, AutoGen, LlamaIndex, Hugging Face
Agentic AI: Multi-Agent Systems, Tool Calling, Model Context Protocol (MCP)
RAG & Search: RAG, Semantic Search, Hybrid Search, Reranking, Embedding Models, Metadata Filtering, RAG Evaluation
Vector DBs: Pinecone, ChromaDB, Weaviate, pgvector, FAISS
Backend: Python, FastAPI, REST APIs, PostgreSQL, SQL
Cloud & DevOps: AWS (S3, Lambda, EC2, Bedrock), Docker, Git, CI/CD
MLOps & Evaluation: LangSmith, RAGAS, TruLens, MLflow

EXPERIENCE

GenAI Engineer — *Jul 2025 – Present*
Prozech, USA

- A production RAG pipeline (LangChain, GPT-4o, Pinecone) with OpenAI embeddings and SQL-indexed metadata powers semantic search across 50K+ enterprise documents, cutting analyst research time by 65% and reaching 91% RAGAS accuracy.
- Designed a LangGraph multi-agent system (evaluated against CrewAI) — a supervisor agent coordinating 4 sub-agents for data extraction, summarization, and report generation, with outputs cross-validated against Claude to cut hallucination risk — handling 300+ tasks daily and reducing manual review overhead by 40%.
- Containerized FastAPI microservices with Docker, deployed on AWS EC2 through a Git-based CI/CD pipeline, exposing MCP-compatible tool-calling and structured JSON outputs while sustaining sub-200ms p95 latency across 1,000+ daily calls.
- Iterated on **prompt engineering strategies** — chain-of-thought, few-shot, role prompting, structured outputs — raising task completion quality scores by 28%, tracked through LangSmith tracing and RAGAS relevance metrics.
- Presented evaluation results to product stakeholders while partnering with engineering to roll out LLMOps practices adopted org-wide — prompt versioning, evaluation pipelines, cost tracking dashboards, and hallucination guardrails via LangSmith and TruLens — cutting incident response time by 50%.

LLMOps QA Engineering — *Jan 2025 – Jun 2025*
Tutorify AI, USA

- Built **LLM evaluation frameworks** with RAGAS and TruLens spanning 500+ test cases, surfacing hallucination patterns and lifting output reliability by 34% via prompt refinement and model tuning.
- Automated **Python testing pipelines** to validate RAG retrieval accuracy, latency, and cost efficiency across 8 LLM models, cutting manual QA effort by 60%.
- Presented evaluation findings to engineering leads and wrote LLMOps documentation and best-practice guides for prompt versioning and incident response that the team adopted as its standard playbook, reducing production issues by 45%.

AI/ML Engineer — *Mar 2022 – Aug 2023*
Anvizon, India

- Automated document tagging and routing with **NLP classification pipelines** in Python and Hugging Face Transformers, cutting manual categorization effort by 70% while processing 10K+ documents weekly.
- End-to-end **ETL and feature engineering pipelines** (Python, PostgreSQL) brought data preparation time down from 8 hours to under 90 minutes per run.
- Fine-tuned a **BERT-based sentiment model** on domain-specific data to an 87% F1 score, then integrated it into a customer feedback dashboard the product and operations teams still use weekly.
- Collaborated with a 4-person cross-functional team to deliver an **automated reporting system** that reduced weekly manual reporting from 6 hours to 20 minutes, directly improving team productivity.

PROJECTS

Agentic RAG Chatbot (LangGraph, GPT-4o, Pinecone, FastAPI, LangSmith)

- Multi-agent RAG system with persistent memory and citation tracking over 20K documents; 89% answer relevance and 94% faithfulness on RAGAS.

EDUCATION

University of New Haven — *Aug 2023 – May 2025*
M.S. Data Science / GPA: 3.74 / Coursework: NLP, Deep Learning, ML, AI, Data Engineering